

T = z

1. $x(t+1) = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} x(t) + \begin{pmatrix} 1 \\ 0 \end{pmatrix} (-2)^t$

Calcoliamo il polinomio caratteristico della matrice A:

$P(s) = \det \begin{pmatrix} 1-s & 1 \\ 0 & -1-s \end{pmatrix} = (1-s)(-1-s) = (s-1)(s+1)$

$\sigma(A) = \{1, -1\}$

A tempo discreto la soluzione generale dell'omogenea è: $X_{g,o}(t) = A^T c$

Ora calcoliamo la matrice A^T.

La matrice A è diagonalizzabile quindi - per definizione - possiamo applicare direttamente la formula:

$A^T = V \Lambda^T V^{-1}$
Matrice diagonale definita come l'autovettore elevato a T.
Inversa di V.
Matrice dove le colonne sono gli autovettori relativi allo spettro di A.

Calcolo degli autovettori di V:

$\lambda_1 = 1$

$\begin{pmatrix} 0 & 1 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} y=0 \\ -2y=0 \end{cases} \rightarrow (x, 0) \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} = v_1$

$\lambda_2 = -1$

$\begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} 2x+y=0 \\ x=-\frac{1}{2}y \end{cases} \rightarrow (-\frac{1}{2}y, y) \rightarrow \begin{pmatrix} -\frac{1}{2} \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} -1 \\ 2 \end{pmatrix} = v_2$

$V = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix} \Rightarrow V^{-1} = \begin{cases} a_{11} = (-1)^2 \det(2) = 2 \\ a_{12} = (-1)^3 \det(0) = 0 \\ a_{21} = (-1)^3 \det(-1) = 1 \\ a_{22} = (-1)^4 \det(1) = 1 \end{cases} \Rightarrow \text{cof} = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}^T$

$V^{-1} = \frac{1}{2} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{pmatrix}$

$A^T = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1^T & 0 \\ 0 & (-1)^T \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1^T & -(-1)^T \\ 0 & 2(-1)^T \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{pmatrix}$
 $= \begin{pmatrix} 1^T & \frac{1}{2} - \frac{(-1)^T}{2} \\ 0 & (-1)^T \end{pmatrix} = \begin{pmatrix} 1 & \frac{1}{2} - \frac{(-1)^T}{2} \\ 0 & (-1)^T \end{pmatrix}$

Pertanto la soluzione generale dell'omogenea è la seguente:

$X_{g,o}(t) = \begin{pmatrix} 1 & \frac{1}{2} - \frac{(-1)^T}{2} \\ 0 & (-1)^T \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} c_1 + \left(\frac{1}{2} - \frac{(-1)^T}{2}\right)c_2 \\ c_2 (-1)^T \end{pmatrix}$

Stiamo parlando di un ingresso esponenziale. Nelle equazioni alle differenze prendevamo come soluzione particolare:

$y_p = a \cdot (exp)$

Pertanto la nostra soluzione particolare sarà di questo tipo:

$y_p(t) = (-2)^T \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$, Per avere risonanza dovevo avere nel vettore generale espressioni di questo tipo: $\begin{pmatrix} -2 \end{pmatrix}^T c$

Non ho risonanza.

Ora questa soluzione particolare deve soddisfare l'equazione differenziale:

$X_p(t+1) = A x_p(t) + B w(t)$

Pertanto:

$$(-2)^{T+1} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = (-2)^T \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + (-2)^T \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\cancel{(-2)}^T (-2) \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \cancel{(-2)}^T \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \cancel{(-2)}^T \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -2a_1 \\ -2a_2 \end{pmatrix} = \begin{pmatrix} a_1 + a_2 + 1 \\ -a_2 \end{pmatrix}$$

$$\begin{cases} a_1 + a_2 + 1 = -2a_1 \\ -a_2 = -2a_2 \end{cases} \rightarrow \begin{cases} 3a_1 = -1 \\ a_2 = 0 \end{cases} \rightarrow \begin{cases} a_1 = -\frac{1}{3} \\ a_2 = 0 \end{cases}$$

$$y_p(T) = (-2)^T \begin{pmatrix} -\frac{1}{3} \\ 0 \end{pmatrix}$$

PERTANTO:

$$X_{g, no}(T) = \begin{bmatrix} 1 & \frac{1}{2} - \frac{(-1)^T}{2} \\ 0 & (-1)^T \end{bmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + (-2)^T \begin{pmatrix} -\frac{1}{3} \\ 0 \end{pmatrix}$$

ORA:

$$X(0) = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} -\frac{1}{3} \\ 0 \end{pmatrix}$$

$$C = X_0 + \begin{pmatrix} \frac{1}{3} \\ 0 \end{pmatrix}$$

$$X_{g, no}(T) = \begin{bmatrix} 1 & \frac{1}{2} - \frac{(-1)^T}{2} \\ 0 & (-1)^T \end{bmatrix} \left(X_0 + \begin{pmatrix} \frac{1}{3} \\ 0 \end{pmatrix} \right) + (-2)^T \begin{pmatrix} -\frac{1}{3} \\ 0 \end{pmatrix}$$

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$$X_{g, no}(T) = \begin{bmatrix} 1 & \frac{1}{2} - \frac{(-1)^T}{2} \\ 0 & (-1)^T \end{bmatrix} X_0 + \begin{bmatrix} 1 & \frac{1}{2} - \frac{(-1)^T}{2} \\ 0 & (-1)^T \end{bmatrix} \begin{pmatrix} \frac{1}{3} \\ 0 \end{pmatrix} + (-2)^T \begin{pmatrix} -\frac{1}{3} \\ 0 \end{pmatrix}$$

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$$2. \quad x(t+1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} x(t) + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \delta_0(t)$$

Calcoliamo il polinomio caratteristico:

$$P(s) = \begin{bmatrix} 1-s & 0 \\ 0 & 1-s \end{bmatrix} = (1-s)^2 = (s-1)^2 \quad \text{LA MATRICE NON È DIAGONALIZZABILE.}$$

PASSO PER LA DECOMPOSIZIONE SPETTRALE

Decompongo in fattori semplici:

$$\frac{1}{(s-1)^2} = \frac{A}{s-1} + \frac{B}{(s-1)^2}$$

$$B = \lim_{s \rightarrow 1} (s-1)^2 \frac{1}{(s-1)^2} = 1$$

$$f_1(s) = \frac{1}{(s-1)^2} \frac{P(s)}{P(s)} = \frac{1}{\cancel{(s-1)^2}} \cdot \frac{\cancel{(s-1)^2}}{(s-1)^2} = \frac{1}{(s-1)^2}$$

$$e_1(s) = 1$$

$$E_1 = e_1(A) = I$$

Ricorda la matrice non è diagonalizzabile quindi non puoi scomporre $A^T = \lambda_1^T E_1 + \dots + \lambda_m^T E_m$

$$D = \lambda_1 E_1 = 1I = I$$

$$N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \emptyset \quad k=1 \text{ indice di nilpotenza}$$

$$A = D + N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \checkmark$$

$$A^T = D^T = \lambda_1^T E_1 = 1^T E_1 = 1 E_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$X_{g,0}(\tau) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

Ipotesi:

$$X_p(\tau) = \phi_0(\tau) \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

Non ho risonanza perché non ho un termine C moltiplicato per l'impulso unitario, quindi la soluzione avrà lo stesso grado dell'ingresso, ossia 0.

Ora:

$$\phi_0(\tau+1) \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \phi_0(\tau) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \begin{pmatrix} 0 \\ \phi_0(\tau) \end{pmatrix}$$

$$0 = \begin{pmatrix} a_1 \phi_0(\tau) \\ a_2 \phi_0(\tau) + \phi_0(\tau) \end{pmatrix}$$

$$\begin{cases} a_1 \phi_0(\tau) = 0 \\ a_2 \phi_0(\tau) + \phi_0(\tau) = 1 \end{cases} \rightarrow \begin{cases} a_1 = 0 \\ a_2 + 1 = 0 \end{cases} \rightarrow \begin{cases} a_1 = 0 \\ a_2 = -1 \end{cases}$$

Per tanto:

$$X_p(\tau) = \phi_0(\tau) \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

Ora:

$$X_{g,0}(\tau) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \phi_0(\tau) \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$X(0) = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$C = X_0 - \begin{pmatrix} 0 \\ -1 \end{pmatrix} = X_0 + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Per tanto

$$X_{g,0}(\tau) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \left(X_0 + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) + \phi_0(\tau) \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

comp. Pseudo TRANSITORIA Pseudo-PERMANENTE

$$X_{g,0}(\tau) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} X_0 + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \phi_0(\tau) \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

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3.

$$x(t+1) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} x(t) + \begin{pmatrix} 1 \\ -1 \end{pmatrix} t$$

perché $q(z) = z^{-1}$, $x_p(t) = z^{-1} T \begin{pmatrix} a_1 \\ \vdots \end{pmatrix}$
 $\downarrow$
 $1^T \cdot T$

il polinomio caratteristico è:

$$\det \left(\begin{bmatrix} -\tau & 0 \\ 0 & 1-\tau \end{bmatrix} \right) = -\tau(1-\tau) = -\tau + \tau^2 \Rightarrow \tau^2 - \tau \rightarrow \tau(\tau-1).$$

$$\lambda_1, \lambda_2 \in \sigma(A) = \{0, 1\} \text{ MATRICE diagonalizzabile.}$$

TROVO AUTORETTORI:

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$A^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1^T \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$X_{g,0}(\tau) = \begin{bmatrix} 0 & 0 \\ 0 & 1^T \end{bmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ c_2 \end{pmatrix} \quad \text{HO RISONANZA.}$$

$MULTIPLO \quad w(\tau) = \begin{pmatrix} \tau \\ -\tau \end{pmatrix}$

Se $c_2=1$
esce 1^T
ossia una parte del sistema

$X_p(\tau) = \begin{matrix} 1^T \tau \\ \vdots \\ 1 \end{matrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \begin{matrix} 1^T \\ \vdots \\ 1 \end{matrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$

AUMENTO IL GRADO DELLA X.P.

$$X_p(\tau) = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \tau \begin{pmatrix} a_3 \\ a_4 \end{pmatrix} + \tau^2 \begin{pmatrix} a_5 \\ a_6 \end{pmatrix}$$

deve soddisfare:

$$X_p(\tau+1) = A X_p(\tau) + B w(\tau)$$

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + (\tau+1) \begin{pmatrix} a_3 \\ a_4 \end{pmatrix} + (\tau+1)^2 \begin{pmatrix} a_5 \\ a_6 \end{pmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \tau \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} a_3 \\ a_4 \end{pmatrix} + \tau^2 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} a_5 \\ a_6 \end{pmatrix} + \begin{pmatrix} \tau \\ -\tau \end{pmatrix}$$

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \begin{pmatrix} a_3\tau + a_3 + a_5(\tau+1)^2 \\ a_4\tau + a_4 + a_6(\tau+1)^2 \end{pmatrix} = \begin{pmatrix} \tau \\ a_2 + a_4\tau + \tau^2 a_6 - \tau \end{pmatrix}$$

$$\begin{pmatrix} a_1 + a_3\tau + a_3 + a_5\tau^2 + a_5 + 2a_5\tau \\ a_2 + a_4\tau + a_4 + a_6\tau^2 + a_6 + 2a_6\tau \end{pmatrix} = \begin{pmatrix} \tau \\ a_2 + a_4\tau + \tau^2 a_6 - \tau \end{pmatrix}$$

$$\begin{cases} a_1 + a_3 + a_5 = 0 \\ a_3 + 2a_5 = 1 \\ a_5 = 0 \\ a_2 + a_4 + a_6 = a_2 \\ a_4 + 2a_6 = a_4 - 1 \\ a_6 = a_6 \end{cases} \rightarrow \begin{cases} a_5 = 0 \\ a_3 = 1 \\ a_1 = -1 \\ a_6 = -\frac{1}{2} \\ a_4 = \frac{1}{2} \\ a_2 = 0 \text{ (ARBITR)} \end{cases} \quad X_p(\tau) = \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \tau \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix} + \tau^2 \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}$$

quindi la:

$$X_{g,w_0}(\tau) = \begin{bmatrix} 0 & 0 \\ 0 & 1^T \end{bmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \tau \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix} + \tau^2 \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}$$

ORA

$$X(0) = \begin{pmatrix} 0 \\ c_2 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \rightarrow X(\tau) = \begin{bmatrix} 0 & 0 \\ 0 & 1^T \end{bmatrix} \left(x_0 + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) + \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \tau \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix} + \tau^2 \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}$$

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$$C = x_0 + \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$X(\tau) = \begin{bmatrix} 0 & 0 \\ 0 & 1^T \end{bmatrix} x_0 + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \tau \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix} + \tau^2 \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}$$

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$$4. \quad x(t+1) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} x(t) + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \left(\frac{1}{2} \right)^t$$

IL POLINOMIO CARATTERISTICO È:

$$\det \begin{pmatrix} 1-s & 1 & 0 \\ 0 & 1-s & 0 \\ 0 & 0 & 1-s \end{pmatrix} = (1-s)^3 = (s-1)^3 \quad \lambda_1 = 1 \quad \text{MATRICE NON DIAGONALIZZABILE}$$

PASSO PER LA DECOMPOSIZIONE SPETTRALE

Decomporre in fattori semplici:

$$\frac{1}{(s-1)^3} = \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{C}{(s-1)^3}$$

$$C = \lim_{s \rightarrow 1} \cancel{(s-1)^3} \cdot \frac{1}{\cancel{(s-1)^3}} = 1$$

$$\lim_{s \rightarrow \infty} \frac{1 \cdot s}{(s-1)^3} = 0 \rightarrow A = 0$$

$$\lim_{s \rightarrow \infty} \frac{As}{(s-1)^3} + \lim_{s \rightarrow \infty} \frac{Bs}{(s-1)^2} + \lim_{s \rightarrow \infty} \frac{Cs}{(s-1)} = A + 0 + 0$$

$$s \rightarrow \infty \quad s-1 \quad s \rightarrow \infty \quad (s-1)^2 \quad s \rightarrow \infty \quad e^{-1/s}$$

sostituendo a $W(s)$ un valore che non sia un polo, $s=2$, $B=0$.

$$f_1(s) = \frac{1}{(s-1)^3} \cdot \frac{P(s)}{P(s)} = \frac{1}{(s-1)^3} \cdot \frac{(s-1)^3}{P(s)} = \frac{1}{P(s)}$$

$$e_1(s) = 1$$

$$E_1 = e_1(A) = I$$

$$D = \lambda_1 E_1 = 1I = I$$

$$N = A - D = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \checkmark \quad A = D + N \quad \checkmark$$

$$N^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \underline{k=2}$$

$$A^T = D^T + \binom{T}{1} D^{T-1} N + \binom{T}{2} D^{T-2} N^2 + \dots + \binom{T}{k-1} D^{T-k+1} N^{k-1}$$

$$\begin{aligned} A^T &= D^T + T D^{T-1} N = \begin{bmatrix} 1^T & 0 & 0 \\ 0 & 1^T & 0 \\ 0 & 0 & 1^T \end{bmatrix} + T \begin{bmatrix} 1^{T-1} & 0 & 0 \\ 0 & 1^{T-1} & 0 \\ 0 & 0 & 1^{T-1} \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1^T & 0 & 0 \\ 0 & 1^T & 0 \\ 0 & 0 & 1^T \end{bmatrix} + \begin{bmatrix} T 1^T & 0 & 0 \\ 0 & T 1^T & 0 \\ 0 & 0 & T 1^T \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1^T & 0 & 0 \\ 0 & 1^T & 0 \\ 0 & 0 & 1^T \end{bmatrix} + \begin{bmatrix} 0 & T 1^T & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1^T & T 1^T & 0 \\ 0 & 1^T & 0 \\ 0 & 0 & 1^T \end{bmatrix} \end{aligned}$$

$$X_{g,o}(T) = \begin{bmatrix} 1^T & T 1^T & 0 \\ 0 & 1^T & 0 \\ 0 & 0 & 1^T \end{bmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 1^T c_1 + T 1^T c_2 \\ 1^T c_2 \\ 1^T c_3 \end{pmatrix}$$

$$X_p(T) = \left(\frac{1}{2}\right)^T \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \quad \text{DENTRO AL VETTORE delle soluzioni non ho } \left(\frac{1}{2}\right)^T \cdot C, \text{ non posso AVERE RISONANZA CON } c_1 = \frac{1}{2}, \text{ perche' sarebbe } \frac{1}{2} \neq \left(\frac{1}{2}\right)^T.$$

NON HO RISONANZA.

Questa deve soddisfare l'equazione differenziale:

$$\begin{aligned} \left(\frac{1}{2}\right)^{T+1} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} &= \left(\frac{1}{2}\right)^T \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \left(\frac{1}{2}\right)^T \end{pmatrix} \\ \cancel{\left(\frac{1}{2}\right)^T} \cdot \frac{1}{2} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} &= \cancel{\left(\frac{1}{2}\right)^T} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \cancel{\left(\frac{1}{2}\right)^T} \end{aligned}$$

$$\begin{pmatrix} \frac{a_1}{2} \\ \frac{a_2}{2} \\ \frac{a_3}{2} \end{pmatrix} = \begin{pmatrix} a_1 + a_2 \\ a_2 \\ a_3 + 1 \end{pmatrix} \quad \begin{cases} a_1 + a_2 = \frac{1}{2} a_1 \\ a_2 = \frac{1}{2} a_2 \\ a_3 + 1 = \frac{1}{2} a_3 \end{cases} \rightarrow \begin{cases} a_1 = 0 \\ a_2 = 0 \\ \frac{a_3}{2} = -1 \end{cases} \rightarrow \begin{cases} a_1 = 0 \\ a_2 = 0 \\ a_3 = -2 \end{cases}$$

$$X_p(T) = \left(\frac{1}{2}\right)^T \begin{pmatrix} 0 \\ 0 \\ -2 \end{pmatrix}$$

$$X_{g,No}(T) = \begin{bmatrix} 1^T & T 1^T & 0 \\ 0 & 1^T & 0 \\ 0 & 0 & 1^T \end{bmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} + \left(\frac{1}{2}\right)^T \begin{pmatrix} 0 \\ 0 \\ -2 \end{pmatrix}$$

ORA:

$$X(0) = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -2 \end{pmatrix}$$

$$r = v = 101$$

$$u = \lambda_0 = \begin{pmatrix} 0 \\ -2 \end{pmatrix}$$

PERTANTO %

$$X(\tau) = \begin{bmatrix} 1^T & \tau 1^T & 0 \\ 0 & 1^T & 0 \\ 0 & 0 & 1^T \end{bmatrix} \left(X_0 - \begin{pmatrix} 0 \\ 0 \\ -2 \end{pmatrix} \right) + \begin{pmatrix} 1 \\ 2 \end{pmatrix}^T \begin{pmatrix} 0 \\ 0 \\ -2 \end{pmatrix}$$

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$$X(\tau) = \begin{bmatrix} 1^T & \tau 1^T & 0 \\ 0 & 1^T & 0 \\ 0 & 0 & 1^T \end{bmatrix} X_0 + \begin{bmatrix} 1^T & \tau 1^T & 0 \\ 0 & 1^T & 0 \\ 0 & 0 & 1^T \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix}^T \begin{pmatrix} 0 \\ 0 \\ -2 \end{pmatrix}$$

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